Software Engineering

BCSE301P Assignment 4

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1. Project Title

MomentsHub: A Collaborative Daily Photo Capsule

2. Problem Statement

In today's hyper connected world, major social media platforms are often overwhelming. They are public facing, driven by algorithms that prioritize engagement over genuine connection, and filled with curated, "perfect" moments rather than authentic, everyday life. This creates a gap for small, private groups of friends and family who want to share simple, daily moments without the pressure and noise of a large social network.

Existing solutions like shared cloud photo albums (e.g., Google Photos) are excellent for storage but lack a focused, engaging experience to encourage daily interaction. They function as archives rather than as living, breathing spaces for connection.

The problem is the absence of a simple, intimate, and lightweight digital space for close-knit groups to collaboratively build a shared history of their everyday lives, one moment at a time.

3. Motivation and Rationale

The motivation for developing MomentsHub is twofold, stemming from both a social need and a technical learning objective.

Social Motivation: In an era where friends and family are often geographically dispersed, maintaining a sense of connection through shared daily experiences is incredibly valuable. I was motivated by the idea of creating something that feels like the digital equivalent of a family scrapbook—a private, safe, and joyful space. The goal is to move away from the performative nature of mainstream social media and build a tool that fosters genuine connection and nostalgia. The "Photo of the Day" feature is specifically designed to spark conversations and bring a small, shared memory to the forefront of everyone's day, strengthening bonds through consistent, low effort interaction.

Technical Motivation: As a third-year computer science student, this project presents the perfect opportunity to gain hands-on experience with a modern, full-stack web development workflow. It allows me and my teammate to move beyond theoretical knowledge and apply our skills to build a real-world, end-to-end application. The

chosen tech stack (React, Firebase, Tailwind CSS) is highly relevant in the current industry. This project provides a practical learning ground for crucial skills, including database design (Firestore), user authentication, cloud file storage, real-time data synchronization, and deploying a scalable application—all of which are essential for a future career in software engineering.

4. Software Development Life Cycle (SDLC) Model

Chosen Model: The Iterative Model

For the MomentsHub project, the most suitable SDLC model is the **Iterative Model**.

Description of the Iterative Model:

The Iterative Model is a process where the project is developed through a series of repeated cycles, or "iterations." Unlike the rigid, sequential Waterfall model, the iterative approach starts with an initial, simplified implementation of a subset of the software requirements. This initial version is then progressively enhanced through successive iterations until the complete system is developed.

Each iteration consists of its own mini-workflow: it includes requirement analysis, design, implementation, and testing, but only for the features planned for that specific cycle. At the end of each iteration, a new, more functional version of the software is produced.

Justification for Choosing the Iterative Model:

The Iterative Model is the ideal choice for the MomentsHub project for several key reasons:

1. **Well-Defined Core Features:** Our project has a clear and stable set of core requirements (user auth, group creation, photo upload). The Iterative model works perfectly here, as we can build these core features in the first one or two iterations to create a functional baseline product quickly.
2. **Ideal for Small Teams and Short Timelines:** As a two-person team with a short timeline, we cannot afford the extensive upfront planning and documentation required by the Waterfall model. The Iterative model allows us to start developing and producing working software very early in the process, which builds momentum and allows for early feedback.
3. **Allows for Progressive Enhancement:** We can build the application in logical, manageable chunks.
    - **Iteration 1:** Focus on the absolute essentials: User signup/login and the ability to create a group.

    - **Iteration 2:** Add the core feature: Photo uploading and viewing them in a basic gallery.
    - **Iteration 3:** Enhance the experience with features like the "Photo of the Day" carousel, favorites, and tags.
    - **Iteration 4:** Focus on polish, notifications, and deployment.
4. **Facilitates Early Testing and Feedback:** At the end of each iteration, we will have a testable, working version of the application. This allows us to identify and fix bugs early, rather than discovering them all at the end. We can also get feedback from each other (or even a few friends) on the new features, allowing for minor adjustments along the way.
5. **Manages Risk Effectively:** By breaking the project into smaller iterations, we reduce the risk of a major failure. If we encounter an unexpected problem with one feature, it only affects one iteration and can be addressed in the next cycle, rather than jeopardizing the entire project timeline.

In summary, the Iterative model provides the perfect balance of structure and flexibility for a project like MomentsHub, enabling our small team to develop a high quality, feature rich application efficiently within our limited timeframe.